Network Protocol

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1. Encoding and Decoding

Every network protocol message sent in the network begins with a four capital letter sequence which matches one of the predefined network commands, presented later in this document. Depending on the command itself, it can be followed by any number of arguments. In case it is necessary information can be encoded by simply preceding a specific splitter which separates each argument. The splitter used for all encoding and decoding work is a string containing 2 chars: %&
In the code the complete data string is usually called “msg”:

<4 letter protocol>%&<1. argument>%&<2. argument>...%&<n. argument>

A received network protocol message then can be decoded by using the string split method.

2. Network Protocol Commands

LGIN - Login

This command signals that a client / new player has connected to the server. It is only sent from client to server.
The data string is formatted as follows: <username>

LOGT - Logout

This command signals that a client wants to disconnect from the server. All the clients who are still connected get a message. This is sent from the client to the server
No data string is required.

NAME - Set Username

This command signals that a client wants to change its name. Needs to be followed by argument "ClientID". It is sent from clients to server. The server checks whether the name is legal.
The data string is formatted as follows: <new username>

HSCR - Highscores

This command shows the player's intention to have a look at the high score list. The server transmits the encoded high score list to the client. If the command is sent from the client to the server, then no data string is required.
The data string for the client is formatted as follows:
<username>:<score>%&<username2>:<score2>…<usernameN>:<scoreN>

MSGE - Messages

This command sends responses from the server to the clients in the form of messages. It is sent from the server to the client.
The data string is formatted as follows: <message>

CELY - Create new a Lobby

This command signals that the client wants to create a new lobby. The server creates the new lobby and adds the client to it. This command is sent from the client to the server and no data string is needed.

JOLY - Join Lobby

This command indicates that a player wants to join an already existing lobby. The name of the lobby is attached in the data string. This command is only sent from client side to server side. The data string needs LobbyID as an argument.

QILY - Leave Lobby

This command signals the player's intention to leave his lobby. The server displays a leaving message and removes the client from the lobby and the player list. This command is only sent from clients to the server.

NEWG - New Game

This command signals the player's intention to start a new game. The server now initiates a new game with every player who is in the same lobby. No data string is needed.

MOVE - Move

This command sends the movement of a chess figure. The data is formatted as follows:
The data string is formatted as follows: <startSquare>/<endSquare>

CHBD - Chessboard

This command is used to update the chessboard after every move. The server then updates the board and returns it to the players. It encodes from the top left square to the bottom right one and uses an integer for empty spaces and the first char of a piece color + chesspiece initial char:
Chesspiece = <piece color char><chesspiece initial char>
Argument encoding: <Nr of empty Squares<Chesspiece>/.../<Chesspiece><Nr of empty Squares>/

SELT - Select

This message signals that a player has selected a specific figure on the board. This command is sent from the client to the server.

SLTD - Selected

This command tells all clients which figure has been selected by the current player. This command is sent from the server to the clients.

TURN - Turn

This command tells all the players that a new turn has begun and who's turn it is. The current player's ID is transmitted as the data.

STAT - Status

This command relays the current status of a lobby chess board. It is sent from server to client.

GOVR - Game Over

This command signals that the game is over. The command is sent from the server to the clients and the attached string contains the information about the game result.

ENDG - End the game

This command is used to end the game. The server calculates which team has more points and returns the result to the players.

DRAW - Game ends in a draw

This command is used when the game ends in a draw. The server returns a message with the points that were achieved by both teams.

WINN - Winner

This command returns the winner of the game after the game has ended. It is sent from the server to the clients. The data string returns the winners and the achieved points.
Argument encoding:
<lobbyID><clientID1><username1>…<clientID4><username4><gameID><playerTurnID><status><playerpoints1>…<playerpoints4> with a “%&” String splitter in between each argument.

UPPL - Update Players

This command is used to update the list that contains all players.
For example: <clientID 1><username 1><clientID 2><username 2>…<clientID n><username n> with a “%&” String splitter in between each argument.

CHAT - Global Chat

This command signals that a client wants to send a message to all the connected clients. The server then transmits this message to all the clients that are in the global chat.

WCHT - Whisper Chat

This command signals that a client wants to send a private message to another client. This chat message is sent from a client to a server and then redirected to the recipient. The message contains the encoded ClientID of the sender as argument.
Argument encoding: <clientID> <message>

LCHT - Lobby Chat

This command signals that a client has sent a message in the lobby chat. The server sends the message only to the players who are connected to the same lobby. The data string contains the message.

SLOB - Show Lobbies

This command shows all available lobbies to the player. The server sends a list of all lobbies as a data string encoded by the encoder.
Argument encoding: <lobbyID><playercount><username1>…<username4>

LBBS - Update the lobbies

This command is used to update the list that contains all lobbies. It is sent from the server to the clients.
Argument encoding: <lobbyID><playercount><username1>…<username4>

SNAM - Show Name

This command is sent from the client to the server. The server then returns the client's username.

SLNA - Show Lobby Name

This command indicates that a client wants to know in which lobby he is currently in. The server sends a data string containing the LobbyID as well as the current players in the lobby.
Argument encoding: <lobbyID><username1>…<username4>.
This command is sent from the client to the server.

SPLY - Show players

This command shows the client how many and which players are already in the lobby. The server returns a data String containing all information.
Argument encoding: <clientID 1><username 1><clientID 2><username 2>…<clientID n><username n> with a "%&" String splitter in between each argument.

SHGM - Show Games

This command indicates that the player wants to see a list of all the games that are currently being played. The server returns a data string containing all the ongoing games.
Argument encoding: <lobbyID><gameIDt><status><username1>…<username4> with a "%&" string splitter in between each argument. Argument arrangement can be repeated for each game.

PSMS - Past Games

This command indicates that the player wants to see a list of all finished games. The server then returns a list with all the finished games.
Argument encoding: <lobbyID><gameIDt><status><username1>…<username4> with a "%&" string splitter in between each argument. Argument arrangement can be repeated for each game.

OPGM - Open Games

This command indicates that the player wants to see a list of all open games ready to be joined. The server then returns a list with all the lobbies that are still waiting for players.
Argument encoding: <lobbyID><playercount><username1>…<username4>

TMER - Timer

This command starts a timer after initiating a game. It is sent from the server to the clients.

PING - Ping

This command is used to ping the server and check whether the connection is still standing.

PONG - Pong

This command is used to pong the client and check whether the connection is still standing.

SONO - Sound

This command is used to tell the client when and which sound files to play. It is sent from the server side based on the current game stance. The data part of the message contains a string that tells the client which sound to play, for example:
“Start”
“Piece captured”
“Check”